



2027 IEEE International Conference on Power, Control and Computing Technologies (ICPC2T 2027)

NIT Raipur, Chhattisgarh, India | 3-5 March 2027

National Institute of Technology Raipur — An Institute of National Importance

SPECIAL SESSION ON AI-Driven Hosting Capacity and Power Quality Analysis for Renewable-Rich Grids

LIST OF TOPICS

1. Machine learning and probabilistic methods for hosting capacity estimation
2. Voltage rise, harmonic distortion, and flicker from high PV and EV penetration
3. Optimal allocation and sizing of distributed generation and energy storage
4. Active network management: dynamic voltage control and reconfiguration
5. Protection coordination challenges from bidirectional power flow
6. Reactive power support and Volt-VAR optimization for capacity enhancement
7. Electric vehicle charging impact on distribution network power quality
8. Interconnection standards and grid codes for distributed energy resources
9. Demand-side flexibility and demand response for capacity enhancement
10. Real-time low-voltage network monitoring and state estimation
11. AI/ML-based fault and anomaly detection in high-DER distribution feeders
12. Case studies and pilot projects on high-penetration distribution networks
13. Optimization algorithms for Maximum Power Point Tracking
14. IoT and data-driven monitoring for renewable energy systems

ORGANIZING CHAIRS



Dr. Debasmita Panda
NIT Warangal



**Dr. Debari Prasad
Mishra**
IIIT Bhubaneswar



Dr. Jayanta Kumar Sahu
Vignan's IIT, Vizag



**Dr. Surender Reddy
Salkuti**
Woosong Univ., Korea

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CONTACT INFORMATION

- ✉ dpanda@nitw.ac.in
- ✉ debani@iiit-bh.ac.in
- ✉ jayanta@vignaniit.edu.in
- ✉ surender@wsu.ac.kr